



Agraja Wijayawardane

Mechatronic Undergraduate at Curtin Colombo

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Summary

Practical and motivated engineering student with hands-on experience in industrial automation, robotics, machine vision, PLC systems, and factory maintenance. Experienced in automation projects, robotic systems, equipment testing, pump and motor diagnostics, and production-line improvements through training at Phoenix Industries and Rockland Distilleries. Strong interest in solving real industrial problems with good teamwork, adaptability, and problem-solving skills.

Experience

Rockland Distilleries

Seethawakapura

Engineering Intern - Engineering Department

Nov 2025 - Feb 2026

- Industrial Maintenance & Automation Exposure:

Gained hands-on experience in industrial maintenance, flood recovery, pump and motor diagnostics, PLC systems, and production-line automation during 13-week training at Rockland Distilleries.

- Flood Recovery & Equipment Recommissioning:

Assisted in inspection, testing, drying, alignment, and recommissioning of flood-affected motors, pumps, conveyors, filtration units, and control panels.

- Pump & Motor Diagnostic Support:

Supported pump and motor maintenance using dial indicator alignment, insulation resistance testing, bearing inspection, vibration checks, and temperature monitoring.

- CIP Process Automation Improvement:

Analyzed CIP cleaning operations and proposed PLC-based automation using flow sensors, temperature feedback, timed cleaning cycles, and automated pump/valve sequencing.

- Labeling Machine Technical Evaluation:

Participated in automated labeling machine evaluation by comparing servo accuracy, sensor performance, PLC compatibility, throughput, and maintainability.

<https://www.rockland.lk/>

Phoenix Industries Ltd.

Makandura

Engineering Trainee – Automation Department

Dec 2024 - March 2025

- Vision-Based Rejection System (Caltex Project)

Designed and implemented a machine vision system with AI-powered cameras and pneumatic actuators to detect and reject defective bottles in real time. Integrated motorised camera alignment for multiple bottle sizes, reducing setup time.

- Automated Bottle Orientation & Labeling System (Phoenix Horana Factory)
Developed an automated system with HikRobot cameras, PLCs, and pneumatic actuators for orientation correction and defect detection. Added cooling control to prevent label issues.
- Takeout Robot Arm End for Injection Molding Machine (IM65)
Designed and manufactured a robotic arm end effector using SolidWorks and CNC machining. Conducted precision measurements for seamless integration.
- Universal Robot UR5 Robotic Arm Integration
Programmed and optimised a six-axis robotic arm for pick-and-place tasks. Modified dispenser design and scripts to handle multiple wheel sizes.
- Energy Monitoring System (Schneider PM2100 with Arduino)
Interfaced PM2100 energy meter with Arduino via RS485/Modbus RTU for real-time monitoring. Implemented SD card logging and resolved hardware/software communication challenges.
- Technical Skills & Broader Exposure
Designed and developed 3D models, PCBs, and CNC-machined components; programmed and integrated PLCs, robotics, and AI-driven vision systems utilizing industrial protocols (RS485/Modbus). Collaborated across procurement, logistics, and quality assurance functions to enhance operational efficiency while demonstrating strong problem-solving, teamwork, and adaptability.

<https://phoenix.lk/>

Education

Curtin University	Undergraduate (Reading)
Mechatronic Engineering	July 2022
Ananda College	Advance Level • 3 Simple Passes
Combined Maths, Physics, Chemistry	2019/ 2020
Ananda College	Ordinary Level • 6 A's, 2 B's, 1 C
2016	

Skills

Leadership



Teamwork



Analytical skills



Interests

Programming

Photography

Swimming

Volunteering

Curtin Colombo Student Guild	2022/ 2023
Curtin Student Ambassador Program	2022/2023
Ananda College Western Orchestra	2019
Ananda College ICT Society	2019

Professional Qualifications

Tertiary and Vocational Education Commission	2023
National Certificate Plumber NVQ 4	
National Institute of Occupational Safety and Health	2023
Foundation Course	
Center for Housing, Planning, and Building	2023
Quantity Surveying Certificate Course NVQ 4	
NIBM	2017
Certificate in Computer Programming	
NIBM	2017
Certificate in Computer Science	
University of West London	2020
Dip. LCM	
Trinity College London	2018
Grade 8 Theory in Music	
Trinity College London	2018
Piano - Grade 6	
ABRSM	2017
Saxophone - Grade 8	
Sri Lanka Life Saving	Since 2018
Life Guard	

References

Bhanuka Dayawansa
Lecturer and Discipline Lead - Mechanical Engineering and Mechatronics Engineering (Curtin Colombo)
0714922698
<https://www.curtincolombo.lk/>

Sachintha Mello
Lead - Engineering (Phoenix)
0741541912
<https://phoenix.lk/>